THE WHOLE TENSOR, EXTERIOR AND SYMMETRIC ALGEBRAS OF A QUASI-COHERENT OR LOCALLY FREE MODULE HAVE THE SAME PROPERTY

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The tensor, exterior and symmetric algebras of a sheaf of modules, as infinite direct sums of the corresponding powers. The author's section on symmetric and exterior powers, in which the counted statement occurs. Identifies the objects whose quasi-coherence or local freeness is asserted.
- The permanence of quasi-coherence and of local freeness under the passage to the tensor powers, on a fixed open neighbourhood. Invoked inside the counted proof at source lines 3698-3703 and attributed to the proof of an earlier lemma of this chapter. Shows that one suitably chosen neighbourhood works for all the powers simultaneously.

Source reference. Original lemma lemma-tensor-algebra-permanence: sources/stacks-modules/pinned-original-source.tex, source lines 3699–3699. Navigation pointer only: the counted text refers to the Stacks item lemma-tensor-algebra-permanence, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules.*

(1) *If $\mathcal{F}$ is quasi-coherent, then so is each $T(\mathcal{F})$, $\wedge(\mathcal{F})$, and $Sym(\mathcal{F})$.*
(2) *If $\mathcal{F}$ is locally free, then so is each $T(\mathcal{F})$, $\wedge(\mathcal{F})$, and $Sym(\mathcal{F})$.*

Proof. It is not true that an infinite direct sum $\bigoplus \mathcal{G}_i$ of locally free modules is locally free, or that an infinite direct sum of quasi-coherent modules is quasi-coherent. The problem is that given a point $x \in X$ the open neighbourhoods U_i of x on which $\mathcal{G}_i$

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becomes free (resp. has a suitable presentation) may have an intersection which is not an open neighbourhood of x. However, in the proof of Lemma lemma-tensor-algebra-permanence we saw that once a suitable open neighbourhood for $\mathcal{F}$ has been chosen, then this open neighbourhood works for each of the sheaves $\mathrm{T}^n(\mathcal{F})$, $\wedge^n(\mathcal{F})$ and $\mathrm{Sym}^n(\mathcal{F})$. The lemma follows. $\square$